

Development of Standardized Models in Kentucky Using AirSage Data

OTDMUG Meeting

Columbus, OH

December 8, 2016



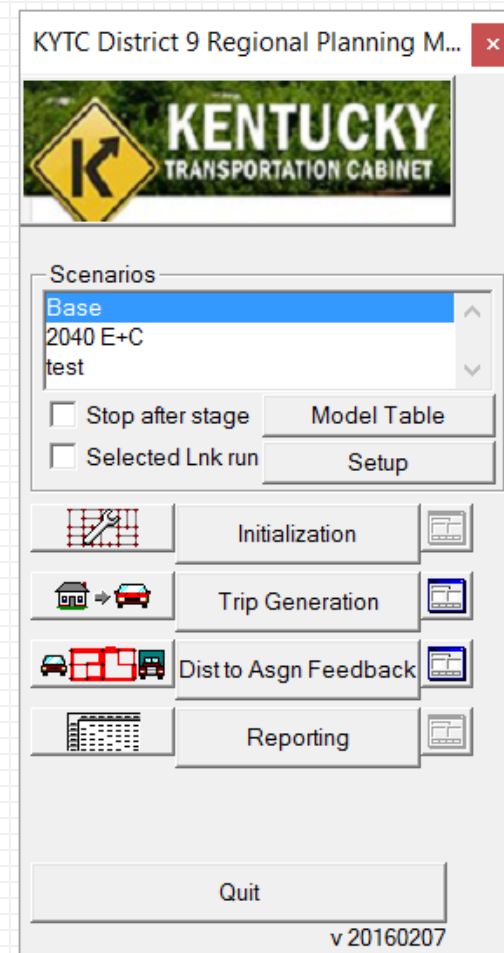
Outline

- **Background**
- **AirSage Pre-processing & Adjustment**
- **AirSage Application**
 - Trip Generation
 - Trip Distribution
 - Time-of-Day
 - Directional Factor
- **Findings**

Background

KYTC's regional TDMs

- Currently using TransCAD 5. In transition to TransCAD 6.
- Standardized organization of files, model stream, script and interface
- Trip-based, 3-step (TG, TD & TA).
- 5 auto trip purposes (HBW, HBO, NHB, HBSchool, HBUniversity).
- 3 truck trip purposes (light, medium, heavy)
- 4 time periods (AM, PM, MD, NT)
- Flexibility of using Big Data to enhance model components.



AirSage in Kentucky

- No recent HH OD survey data in KY. No NHTS add-on.
- Cell phone data (not GPS data).
- Lower cost compared to traditional HH travel surveys.
- Quick turnover.
- AirSage provides ODs by purpose and time period.
- When data collection area is large enough, external trip info. can be obtained. Good data at the edge of study area.
- KYTC first purchased AirSage data for Lexington Model (2012). Complete 5 regional models so far. All used AirSage data.
- KYTC and its consultants have gained thorough insight in using AirSage data for model development.

Background

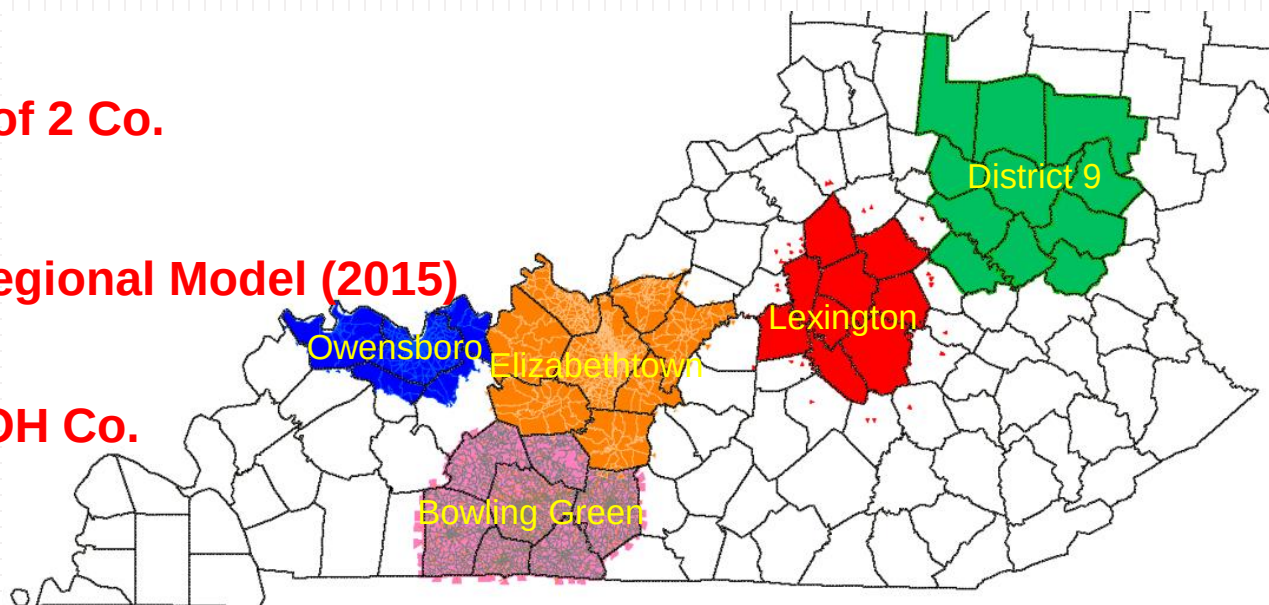
KYTC's Completed Regional Models

- Lexington/Central KY Region (2012)
- Bowling Green/Warren Co. (2014)
- E-town/Hardin Co./Meade Co. (2014)
- Owensboro/Daviess Co. Region (2014)

- 160K pop
- 3 Co. + parts of 2 Co.
- Small MPO

- KYTC District 9 Regional Model (2015)

- 300K pop
- 8 KY Co. + 3 OH Co.
- Rural



AirSage Data Pre-processing

Example of AirSage Data (.CSV)

Origin_Zone	Destination_Zone	Start_Date	End_Date	Aggregation	Subscriber_Class	Purpose	Time_of_Day	Count
74	286	20150415	20150514	WD	Resident	OH	H1800:H2400	12.47
315	225	20150415	20150514	WD	Visitor	OO	H0900:H1500	5.9
432	433	20150415	20150514	WD	Resident	WH	H0900:H1500	11.84

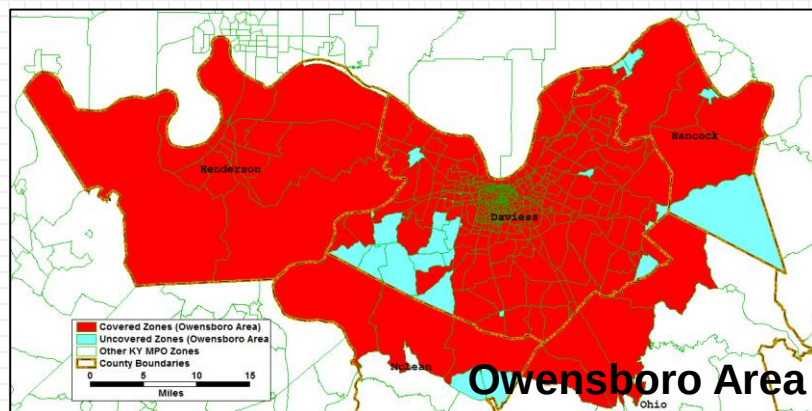
- Must provide pre-defined TAZ polygons to AirSage. AirSage has resolution limit (0.25 square mile grid).
- AirSage provided “expanded” data which matches census population by carriers.
- When multiple carriers exist (e.g., Owensboro Model), averaging data sets minimizes data bias.

AirSage Data Pre-processing

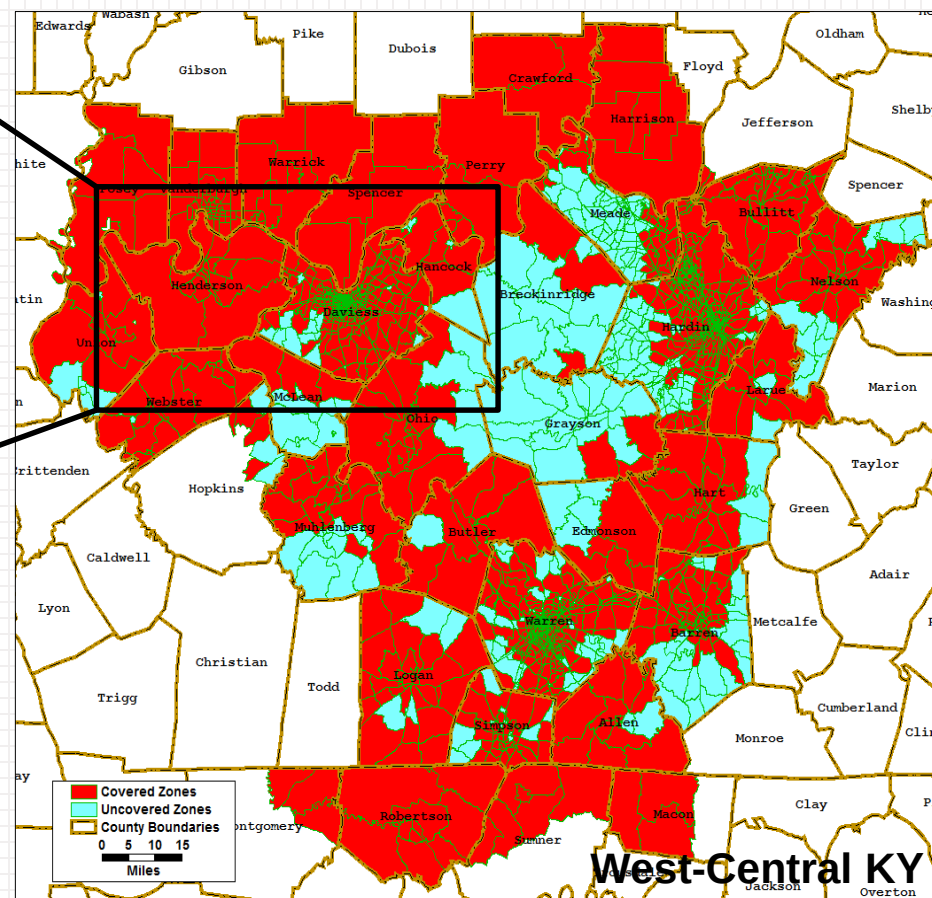
- Day Parts (can be customized to match model)
 - ❑ AM peak = 6:00-9:00
 - ❑ Mid-day = 9:00-15:00
 - ❑ PM peak = 15:00-18:00
 - ❑ Night = 18:00-24:00 & 0:00-6:00
- Trip Purpose: (H-home, W-work, O=other)
 - ❑ HBW = HW, WH
 - ❑ HBO = HO, OH, HH
 - ❑ NHB = WO, OW, WW, OO
- Convert .CSV file to OD matrix (3 purposes x 4 periods)

AirSage Data Coverage Assessment

Owensboro/Daviess Co.

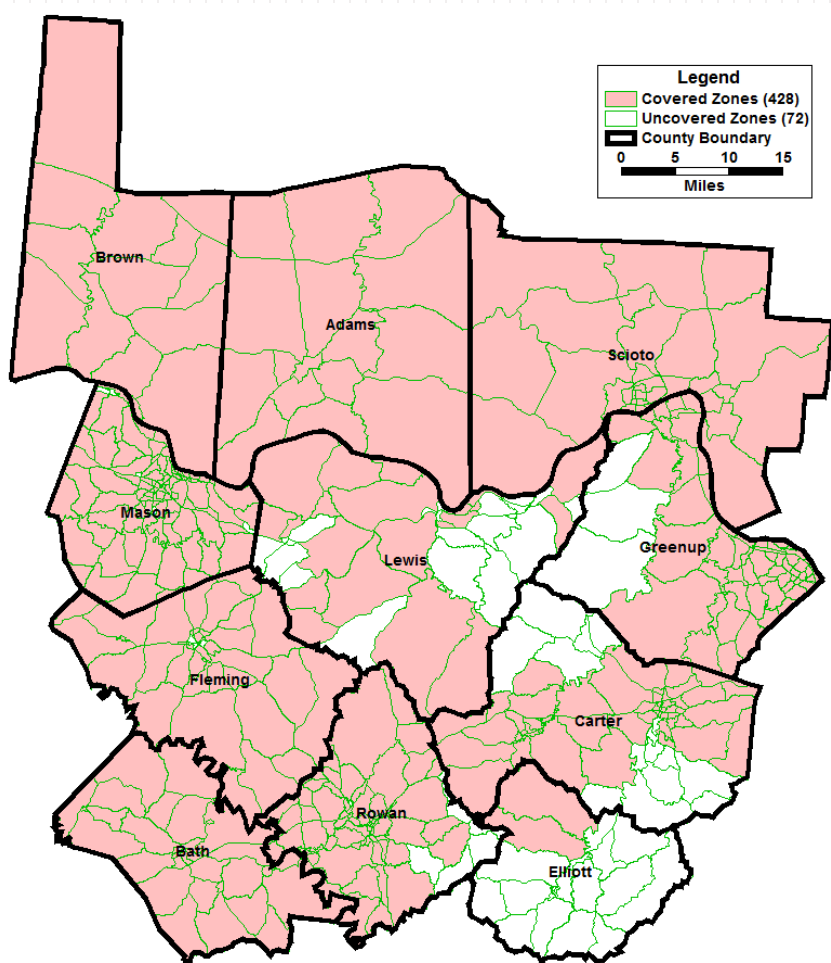


- Data collected for west-central KY
- 95% of 566 Owensboro zones had data
- 28,085 OD pairs had data
- 94% Pop & 97% Emp are covered
- Lack of data in adjacent rural counties. No impact on external trips (0.2% of Owensboro work flows - CTPP).



AirSage Data Coverage Assessment

KYTC District 9 Area



- Data collected only for D9 area
- A 500-zone structure was used for AirSage data acquisition.
- **85%** of 500 zones had data.
- **22,111** OD pairs had data.
- **92% Pop & 93% Emp** are covered
- Uncovered zones are very rural, sparsely populated areas
- A refined 620 zones (final model TAZ) was used for friction factor development.

AirSage Data Adjustments (Intrazonals)

Owensboro

Purpose	Total	Intrazonal	Intrazonal %
HBW	150,012	1,951	1.3%
HBO	330,101	89,822	27.2%
NHB	95,250	29,640	31.1%
Total	575,363	121,413	21.1%

Note: Data is from raw AirSage (all trips)

District 9

Purpose	Total	Intrazonal	Intrazonal %
HBW	118,429	5,339	4.5%
HBO	654,781	324,717	49.6%
NHB	355,881	109,520	30.8%
Total	1,129,090	439,576	38.9%

Note: Data is from raw AirSage (all trips)

- Except for HBW trips, intrazonal trip percentages seem too high
- HBO & NHB intrazonals were reduced to more conventional levels before developing friction factors.

Trip Generation - Trip Rates

- Previous experience indicates AirSage may not be suitable for directly determining trip generation rates. Analysis is limited due to aggregated data & resolution limits.
- Owensboro (small MPO) has too few NHB trips. Home-based additional trips or short-distance trips may not be captured.
- D9 region (rural area) generally has more trips.
- Not much improvement by ODME

Owensboro

Purpose	AirSage		Calibrated Model	
	Internal P's	%	Internal P's	%
HBW	104,628	24%	100,543	20%
HBO	262,339	60%	262,892	52%
NHB	72,025	16%	140,013	28%
Total	438,992	100%	503,448	100%
Trip Rates	6.8		7.7	

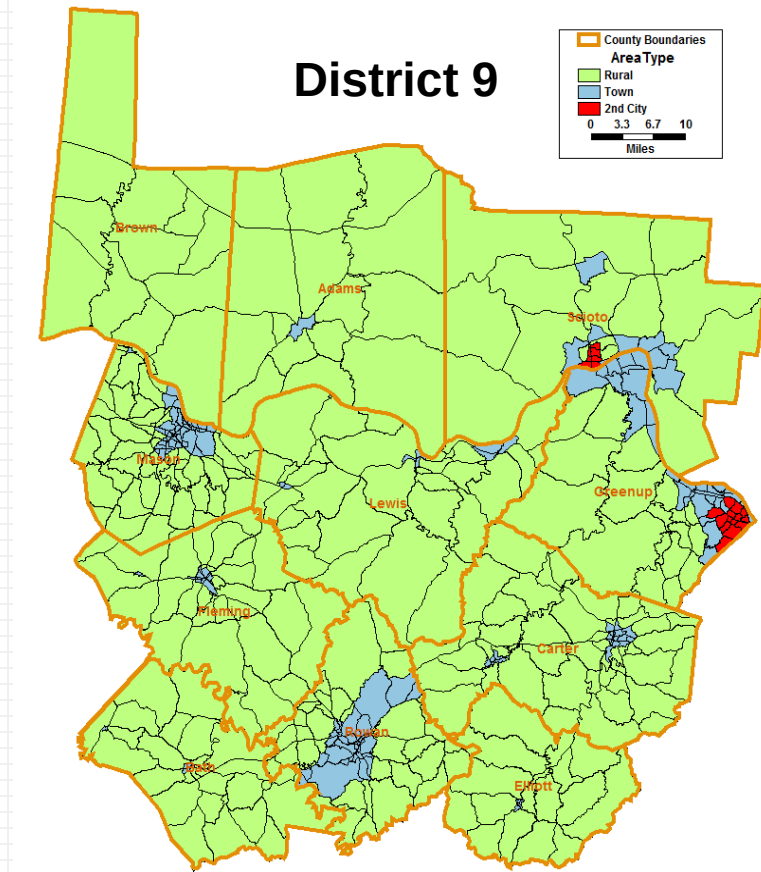
District 9

Purpose	AirSage		Calibrated Model	
	Internal P's	%	Internal P's	%
HBW	118,429	10%	160,704	18%
HBO	654,781	58%	466,274	53%
NHB	355,881	32%	258,846	29%
Total	1,129,090	100%	885,824	100%
Trip Rates	9.6		7.6	

Trip Generation - Area Type Factors

Assumption: Data bias is diminished, if not completely offset between area type samples, by a factoring process.

- Rural, Town & 2nd City
- Convert OD to PA
 - ❑ $HBW = HW + \text{Transpose}(WH)$
 - ❑ $HBO = HO + \text{Transpose}(OH) + HH$
- NHB data not used (not real P's by HH)
- AirSage P's rates (by purpose & area type)
 - ❑ Remove outliers (95% CI)



Trip Generation - Area Type Factors

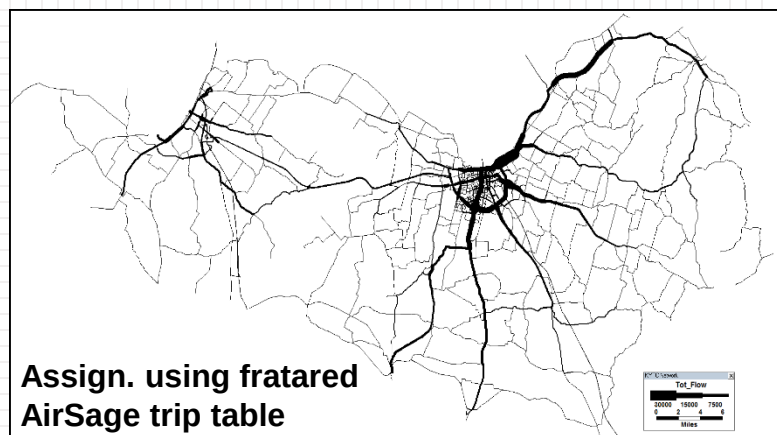
- Area Type Factors (**t-test** of P rates between area types)
 - If significantly different, factor = ratio of P rates
 - Otherwise, factor = 1.0
- Factors of Rural vs. Towns are reasonable (**HBW=1.0, HBO=0.9**)
- Factors for Rural vs. 2nd City, Towns vs. 2nd City are too large.
- Available data (KYSTM) suggests same rates for Towns & 2nd City (consolidated as Non-Rural).
- For NHB, factor (**=1.0**) is derived from NCHRP 716 and KYSTM.

Purpose	D9 Model Area Type Factor	
	Rural	Non-Rural
HBW	1.0	1.0
HBO	0.9	1.0
NHB	1.0	1.0

Trip Distribution – Friction Factors

Owensboro Method

- AirSage PA tables (internal, by purpose) are fratared to match default target P's (NCHRP 716) and A's (NCHRP 365).
- Fratared trip tables work better than raw data



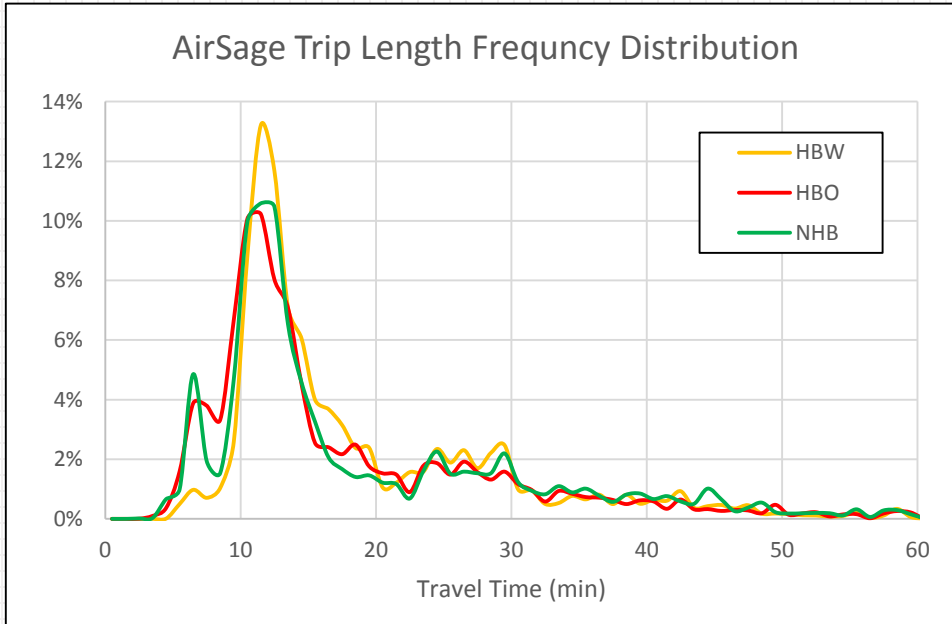
Traffic Assignment	Total Traffic (Count Links)	Avg. Dev % from Counts *
AirSage (raw)	907,259	-59.2%
AirSage (fratared)	1,397,946	-19.8%
Traffic Counts	2,275,196	

* Simple average error, not RMSE

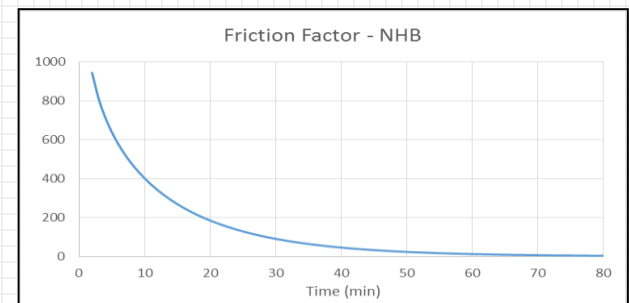
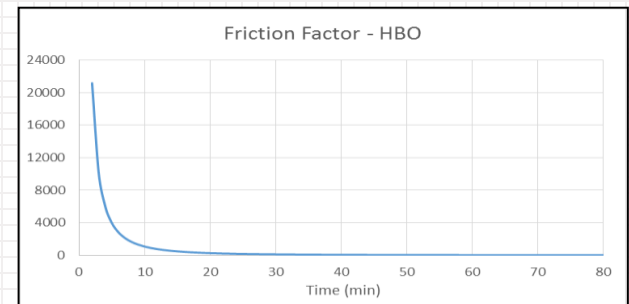
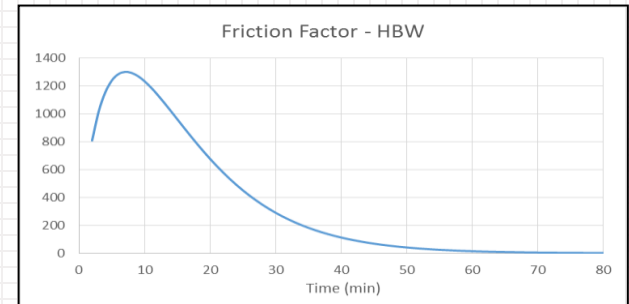
- Friction factors were developed using TransCAD gravity model calibration function. Further adjustment in model calibration.

Trip Distribution – Friction Factors

Owensboro Method



Trip Purpose	Mean Trip Length (min)	
	AirSage	Calibrated Model
HBW	19.1	15.7
HBO	17.8	14.9
NHB	19.4	14.4



Trip Distribution – Friction Factors

District 9 Method (revised approach)

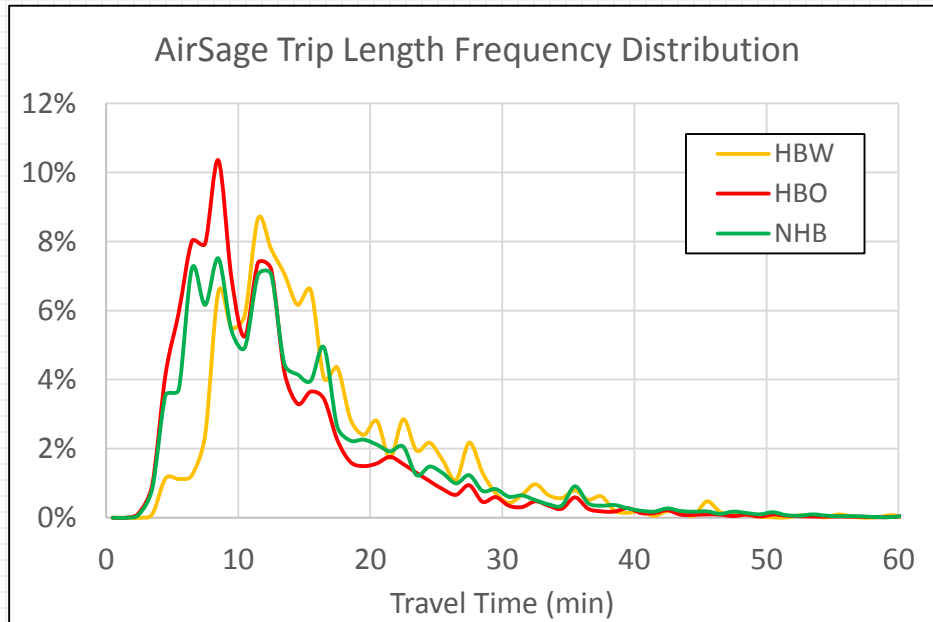
- KY local trip rates usually are unknown and differ from national defaults. Calibrate AirSage OD tables to traffic counts.
- ODME

$$\text{Full seed OD table} = \underset{\substack{\uparrow \\ \text{AirSage} \\ \text{(all purpose)}}}{\text{I-I}} + \underset{\substack{\uparrow \\ \text{Land Use Data} \\ \text{+ NCHRP 716}}}{\text{I-E/E-I}} + \underset{\substack{\uparrow \\ \text{KYSTM extraction} \\ \text{+ Fratar}}}{\text{E-E}}$$

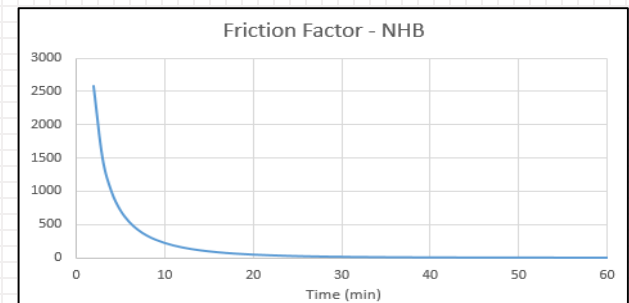
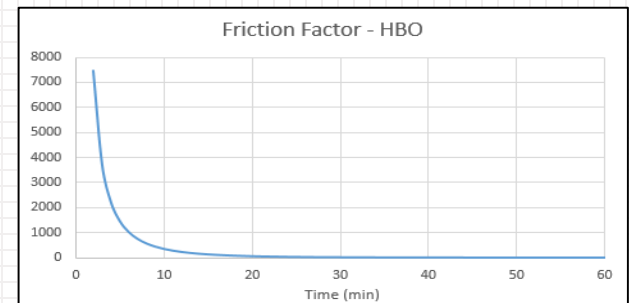
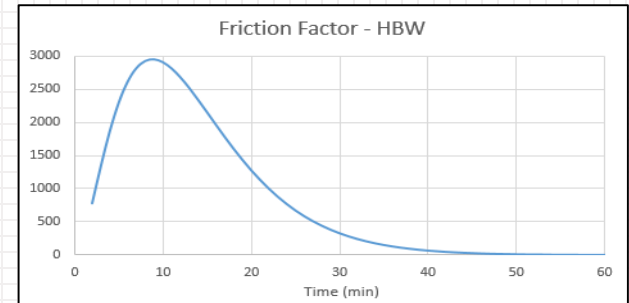
- Disaggregate ODME trip table by purpose – based on original AirSage data
- Friction factors (by purpose) were developed using TransCAD gravity model calibration function.

AirSage – Trip Length Distribution

District 9 Method (revised approach)



Trip Purpose	Mean Trip Length (min)	
	AirSage	Calibrated Model
HBW	16.8	17.3
HBO	13.3	12.5
NHB	15.3	13.7



AirSage TOD & Directional Factors

Owensboro (TOD)

Purpose	AM (6-9am)	Mid-Day (9am-3pm)	PM (3-6pm)	Night (6pm-6am)
HBW	0.26	0.25	0.18	0.31
HBO	0.17	0.30	0.19	0.34
NHB	0.12	0.47	0.23	0.18
E-I Auto	0.20	0.30	0.20	0.30
E-E *	0.19	0.31	0.20	0.30

* Assumed by average of all purposes

District 9 (TOD)

Purpose	AM (6-9am)	Mid-Day (9am-3pm)	PM (3-6pm)	Night (6pm-6am)
HBW	0.28	0.27	0.19	0.26
HBO	0.17	0.30	0.19	0.34
NHB	0.19	0.37	0.22	0.22
E-E *	0.19	0.32	0.20	0.29

* Assumed by average of all purposes

Owensboro (direction – P to A)

Purpose	AM (6-9am)	Mid-day (9am-3pm)	PM (3-6pm)	Night (6pm-6am)
HBW	0.91	0.51	0.16	0.43
HBO	0.85	0.53	0.43	0.33
E-I Auto *	0.85	0.53	0.43	0.33

* Assume equal to HBO

District 9 (direction – P to A)

Purpose	AM (6-9am)	Mid-day (9am-3pm)	PM (3-6pm)	Night (6pm-6am)
HBW	0.94	0.49	0.16	0.39
HBO	0.85	0.54	0.42	0.34
E-I Auto *	0.85	0.54	0.42	0.34

* Assume equal to HBO

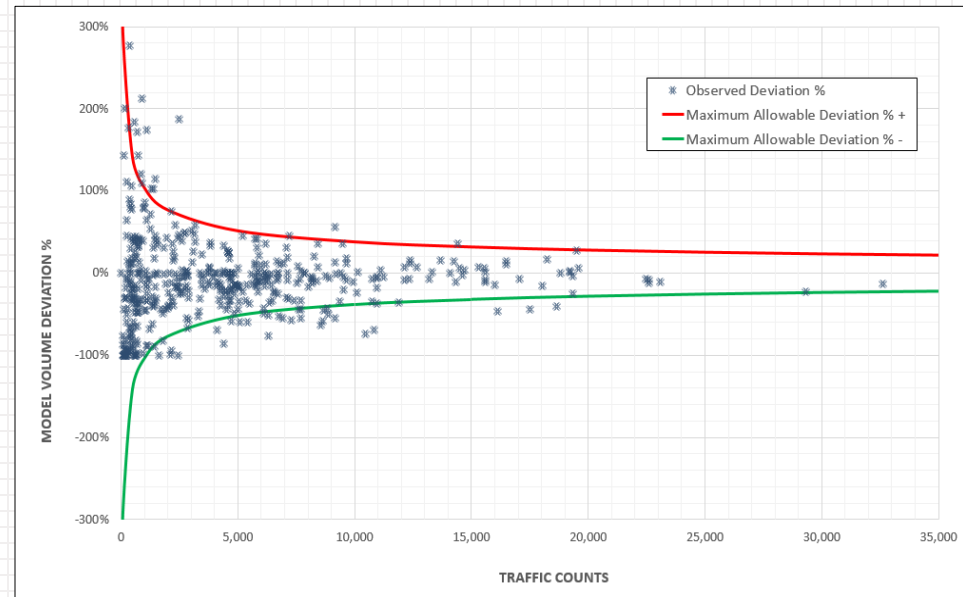
- **AirSage TOD / Directional factors seem reasonable.**
- **TOD factors can be adjusted further to match TOD counts in model calibration.**
- **Large data collection area = good data for external trips & at study area edges.**

Traffic Assignment

AADT Group	RMSE%	
	D9 Model	KYTC Target
0 - 2,000	98.5%	> 55%
2,000 - 5,000	40.8%	45 - 55%
5,000 - 10,000	29.9%	35 - 45%
10,000 - 20,000	17.9%	27 - 35%
20,000 - 30,000	13.4%	24 - 27%
All	42.6%	< 55%

Volume Group	RMSE %				
	AM Peak	Mid Day	PM Peak	Night	KYTC Target
0 - 2,000	55.0%	62.8%	53.5%	58.5%	> 55%
2,000 - 5,000	36.5%	28.9%	23.2%	25.2%	45 - 55%
5,000 - 10,000	n/a	14.2%	n/a	n/a	35 - 45%
Model / Count VMT	1.010	0.955	0.938	0.963	

Functional Class	RMSE%	
	D9 Model	KYTC Target
Interstate	16.2%	< 20%
Other Frwy / Expwy	5.1%	< 20%
Principal Arterial	28.4%	< 30%
Minor Arterials	28.8%	< 40%
Collectors	63.3%	< 70%
Local Roads	127.7%	n/a



AirSage Advantages/Disadvantages

■ Advantages

- ☐ Very large dataset.
- ☐ Low cost when compared to surveys. Quick turnover.
- ☐ Trip purpose, TOD & direction are available
- ☐ Large coverage are readily available.
- ☐ Good data penetration / saturation

■ Disadvantages – **adjustments required**

- ☐ Everything is aggregate. Characteristics of travelers are not available.
- ☐ Trip purpose is based on activity clusters and times of data transmissions.
- ☐ Data transmission times may not accurately reflect when travel occurs.
- ☐ Resolution limits & thresholds of determining device's location (5 min / 300 meters) – may miss short- dist./duration trips, i.e., small zones or urban areas.
- ☐ Unusual results in the trip tables (intrazonal, trip rates)

Summary of Findings

- Findings are based on our understanding of the AirSage product. AirSage product is evolving, so future data may address the issues.
- AirSage data may not be suitable for directly determining trip generation rates. Analysis are limited due to resolution limit and aggregate data.
- AirSage seems to offer a way of estimating the area type impact on trip generation. Validation with other data sources are required.
- Refinement & adjustment of AirSage trip tables are essential before trip distribution analysis. An ODME process is recommended.
- AirSage seems to provide reasonable estimates of existing temporal distribution and directional patterns.
- AirSage data provides a good-starting point for estimating model parameters in small MPO and rural areas.

Questions ?

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